

The Voynich Manuscript as a Writing System

Reproducible measurements against natural languages, medieval scribes,
historical ciphers and text generators

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Abstract

The Voynich manuscript (Yale, Beinecke MS 408) is written in a script that has never been read. We do not propose a decipherment. Instead, we describe *how* the text is written, using more than six hundred preregistered, reproducible experiments, each first validated on synthetic texts in which the effect is known to be present or absent, and replicated on three independent transliterations. We then compare every property with the same measurement on about one hundred natural languages, eight constructed languages, hand-written gibberish produced by volunteers, published Voynich-like generators and ciphers, the eighteenth-century Copiale cipher, and—for the first time in a systematic way—79 real medieval scribes: six Nordic manuscripts (c. 1200–1550) and 73 German manuscripts (1350–1500) transcribed at the level of letter forms, with their original lines and pages.

New findings. (1) The text behaves as a *chain of glyphs* in which the space is a weak boundary: spaces are predictable from the two surrounding glyphs, re-cutting the chain at other permitted points mostly yields words attested elsewhere, and the rules linking the end of one word to the start of the next mirror the rules inside words more than in any of 71 languages. (2) Adjacent words are linked by *junction rules* (*qo-* after *-y/-o/-d*, *o-* after *-n/-r/-s/-m*; *-l/-r* chosen by the next word), applied while writing and looking exactly one word ahead. (3) The *line is a closed unit*: the junction stops at the line break and at the gap left by a drawing, and line starts and ends have their own forms. (4) The scribe *copies from the line immediately above*, adapting the copies to the junction rules. (5) Each line *avoids starting like the line above*. (6) Binary choices between variant forms (*k/t*, *sh/ch*, *-ey/-dy*) follow a *short-lived state* that persists for two to three words regardless of which word is written; it is not copying, not position, separate for each choice, and identical across scribal hands—a rule of the system. (7) Marked variants *drift* towards simpler forms along the line. Two further properties are quantified anew: immediate repetition of the previous word (1.0% of adjacent pairs; 3.9% with one glyph changed) and a strong page, bifolium and paragraph structure.

Comparisons. Junction rules, left-margin avoidance, line-edge forms and immediate repetition have no counterpart in the 79 scribes (they are absent or four to a hundred times weaker). The short state and the in-line drift were not found in the six Nordic scribes (the measurement on the 73 German scribes is in progress), and the Copiale copyist does the opposite of the Voynich, rotating homophones between adjacent words. A slower state crossing line breaks, by contrast, is normal in real scribes. No substitution reading in tens of languages, none of the published readings we could test (Bax, Vatne, Cheshire, Schechter, Gatta), none of the simulated historical mechanisms (homophone choice, Alberti’s index letters, Bacon’s bilateral cipher, Trithemius’ word tables, a line-restarting autokey), no constructed language and no published generator reproduces these properties together.

Conclusion. The Voynich text is the product of a writing system with rules between neighbouring words and at line edges far stronger than those of any scribe, language or generator we tested. These rules do not reveal what the text says, but they exclude many explanations and provide a checklist that any future proposal must reproduce.

Keywords: Voynich manuscript; writing systems; medieval scribes; historical cryptography; quantitative palaeography; preregistration.

Key findings

The first seven findings are, to our knowledge, **new**; the last three strengthen earlier ideas with new measurements.

1. **The text is a chain of glyphs in which the space is a weak boundary.** The rules linking the last glyph of a word to the first glyph of the next mirror the rules between two glyphs inside a word more than in any of 71 comparison languages. Spaces can be predicted from the two surrounding glyphs (accuracy 0.86, above 97% of languages), and cutting the text at other points allowed by the rules mostly yields words attested elsewhere in the manuscript (50%, against 6.5% in languages).
2. **Adjacent words are linked by junction rules.** *qo-* follows words ending in *-y*, *-o*, *-d*; *o-* follows *-n*, *-r*, *-s*, *-m*. A final *-l* or *-r* depends on how the next word begins. The scribe *knows the next word* while finishing the current one. The link is at least four times stronger than in any medieval scribe we tested.
3. **The line is a closed unit.** The link between words stops at the line break and also where a drawing interrupts the line; line starts and ends have their own forms (at line end *-m* replaces *-r*; at line start *s-* and *y-* appear).
4. **The scribe copies from the line immediately above.** Words reproduce, more than chance, words or pieces of line from the line just written, not from lines further up; the copy is adapted to the junction rules with its new neighbour.
5. **Each line avoids starting like the line above.** If the line above begins with *qo-*, the line below begins with *o-*. None of 79 medieval scribes does this: not a single case in 72 tests on the six Nordic scribes, and only three of 73 German manuscripts avoid one single initial, as expected by chance.
6. **Choices between variant forms follow a “state” lasting a few words.** Between the forms *k/t*, *sh/ch*, *-ey/-dy*, the scribe tends to repeat the same choice for two or three words, *whatever word is being written*. It is not copying of similar words, not position in the line, it is separate for each choice and identical in all the hands of the manuscript: a rule of the writing system. We did not find it in the six Nordic scribes nor in the Copiale copyist (the test on the 73 German scribes is in progress).
7. **Marked variants drift along the line** (*qo*, *k*, *sh*, *-ey* become rarer from left to right), three to fifteen times more than in the six Nordic scribes (German scribes in progress).
8. **The Voynich repeats the previous word immediately** (1.0% of adjacent pairs) or a nearly identical one (3.9%), more than every language and every one of the 85 handwritten texts we tested (usually 10 to 100 times more), while having a vocabulary of normal size.
9. **The book has a writing structure:** each page has a strong identity, the bifolium is a working unit (one Currier “language” and almost always one hand), the paragraph opens with a large sign (*p*) and has first and last lines with their own register; *-ey* increases down the page.
10. **No proposed reading survives the controls:** neither ours (substitutions in dozens of languages, solvers, verbose ciphers, nomenclators, content-guided searches) nor the published

ones we could test (Bax, Vatne, Cheshire, Schechter, Gatta), nor the historical cipher mechanisms we simulated (homophones, Alberti's index, Bacon's bilateral cipher, Trithemius' word lists, autokey).

1. Introduction

1.1 The manuscript

The Voynich manuscript is a codex of about 240 pages, with drawings of plants, astronomical and zodiacal diagrams, female figures in pools and pipes, pharmacy jars, and a long final section of text only, divided into short paragraphs marked by stars (usually called “recipes”). Its vellum has been radiocarbon-dated to 1404–1438. The script is neat and confident: it does not look like the work of someone inventing signs while writing. In the 1970s Prescott Currier noticed that the pages fall into two varieties of writing, now called **language A** and **language B**, and that more than one hand wrote the manuscript; Lisa Fagin Davis later distinguished five scribes.

The text is studied through **transliterations** that render each sign with a conventional Latin letter. The most used is the EVA alphabet: a word such as *qokeedy* or *daiin* says nothing about sound; it is only a label for a sequence of signs. Some signs are written in EVA with more than one letter (*ch*, *sh*, the tall “gallows” *k*, *t*, *p*, *f* and their compounds); in all our measurements we count them as a single sign.

1.2 The problem

Over more than a century dozens of readings have been proposed: abbreviated Latin, Romance languages, Hebrew, Old Turkic, Asian languages, ciphers of various kinds. None has convinced the scholarly community, for a simple reason: it is easy to find in a long text a few words that “look like” something, and hard to show that a reading does better than chance. On the other side, those who argue that the text has no meaning must explain why it is so regular.

1.3 What this work offers

Not a new reading, but a set of **measurements with controls**:

- each measurement was **written down before** looking at the result on the Voynich (preregistration);
- each method was **first tested on synthetic texts** in which the effect was present or absent, to check that it recovers it;
- each property of the Voynich was **compared** with the same measurement on languages, scribes, ciphers and generators;
- when an intermediate measurement proved wrong, **we wrote it down** and corrected it (the internal checks are listed in Appendix B).

The result is a description of the Voynich as a **writing system**: what the writer does, word after word and line after line. This description is useful in two ways: it says which explanations are excluded, and it provides a list of properties that any future explanation will have to reproduce.

2. Data

2.1 The Voynich

Three independent transliterations in René Zandbergen’s IVTFF format:

transliteration	alphabet	role
Zandbergen–Landini (ZL)	EVA	reference
Takahashi (IT)	basic EVA	replication
Glen Claston (GC)	v101 (a different segmentation of the signs)	replication

We use the paragraph text, about 35,000 legible words. Labels of drawings, circular texts and radii enter only some tests, which are indicated. Words with illegible signs (0.6%) are discarded. Every main result was replicated on at least two transliterations, most on all three.

2.2 Comparison texts

- **Natural languages:** the Bible in about 100 languages; 71 texts from the Gaskell and Bown (2022) collection, as whole words; Latin technical texts (Pliny, Varro, Isidore, Vitruvius) and an Italian one (Alberti’s *Della Pittura*), to control for genre.
- **Constructed languages:** Esperanto, Volapük, Neo-Quenya, Interlingua, Lojban, Klingon, LOLCat, Toki Pona.
- **Handwritten gibberish:** 38 texts produced by volunteers asked to write an invented text by hand (Gaskell and Bown 2022).
- **Published generators and ciphers that imitate the Voynich:** the Naibbe cipher (Greshko 2025), which encrypts a real Latin or Italian text with a procedure executable by hand; the “self-citation” generator of Timm and Schinner (2020), which produces text without a message by copying and modifying words already written; two amateur generators (U2, U3).
- **Real medieval scribes, at the level of the facsimile** (that is, with letter forms as in the manuscript, abbreviations, and the original lines and pages):
 - six manuscripts from the **Menota** archive (Medieval Nordic Text Archive): Icelandic, Norwegian and one Swedish, from about 1200 to 1550;
 - 73 German manuscripts from the **Referenzkorpus Frühneuhochdeutsch** (ReF), from 1350 to 1500, that is, from the area and century of the Voynich, with lines of 5–8 words like its own;
 - the Anglo-Saxon scribe of the *Hatton Gospels* (from an edition, without original lines).
- **A real historical cipher:** the **Copiale**, an eighteenth-century German enciphered manuscript (105 pages), with the transcription and decipherment of Knight, Megyesi and Schaefer (2011). It uses a homophonic cipher: the same letter can be written with different symbols, at the copyist’s choice.
- **Handwritten medieval technical texts:** from the ReF, a *Book of Nature*, a surgical recipe book, a book of lunar astrology, a *Naturlehre* and a book of magic and alchemy.

All data were taken at fixed versions, with the licences listed in Appendix D. Protected texts are not redistributed.

3. Methods

3.1 The rules we set ourselves

1. **Preregistration.** For each experiment, before running it on the Voynich, we wrote in a dated file the question, the measure, the data and the **criterion** by which we would read the outcome (“if the value exceeds X we will say Y ”).
2. **Testing the code on synthetic texts.** Every measure was first tested on texts built for the purpose, with the effect present and absent. A measure that did not distinguish the two cases was not used: in three cases the experiment was not run and was declared as a gap (Section 11).
3. **Positive and negative controls.** For decipherment tests: a known text enciphered in the same way (which the method must solve) and a text in another language (which shows what is obtained by chance).
4. **Honest intervals.** Uncertainty intervals are computed by resampling whole pages, because neighbouring words are not independent.
5. **A failed step is repeated once**, then the gap is declared.
6. **Corrections are written down.** The laboratory notebook was written only by adding, never by deleting.

3.2 The main measures, in plain words

- **Junction:** how much information the last glyph of a word gives about the first glyph of the next word, beyond what glyph frequencies alone would give (in bits).
- **Junction rule:** how much a choice at the start (or end) of a word depends on the glyph of the neighbouring word, for the same word.
- **Agreement of choices (the “state”):** for a choice between two variants (for example k or t), how much two nearby words of the same line make the same choice more than expected, where the expectation accounts for the preferences of each word, of the page and of the position in the line. It is expressed as a fraction of the maximum possible agreement ($0 =$ chance, positive values = more agreement than chance), like Cohen’s kappa.
- **Copying from the line above:** how many words have an identical or nearly identical word in the line above, beyond chance.
- **Left margin:** how often a line begins like the line above, compared with shuffled lines.

3.3 A decisive methodological correction

Midway through the work we found that the measure of agreement between choices had a **bias** that varied from text to text: in some texts it inflated agreement by up to $+0.37$, in others it lowered it. The cause is that the expectation is estimated from the same data. The correction consists in subtracting the value obtained by shuffling the choices among occurrences of the same word, which has the same bias but no real agreement. All agreement measures reported here are corrected in this way. The correction changed three provisional indications of the work: the Anglo-Saxon scribe does not have the “window” a first measure had attributed to him; Volapük does not resemble the Voynich; and the Voynich does not lie *outside* the range of languages for the shape of its agreement, but at its top.

4. The text as a chain of glyphs: words, spaces and lines

4.1 A fingerprint that no language has

The first fact, partly known, is that the Voynich text is too regular. The uncertainty about the next glyph (a classical measure, called h_2) is 2.2 bits, against 2.6–3.3 in languages with an alphabet of similar size. Taking nine measures together (predictability of glyphs, spaces, vocabulary, repetitions, similarity between words of the same page, links between neighbouring words), the Voynich lies farther from the nearest language than the most isolated language of the sample lies from its own neighbour (distance 9.9 against 4.1). The languages that “most resemble” it (Potawatomi, Chinantec, Ewe, Malagasy) have nothing in common with one another: the Voynich has no natural linguistic family.

Other properties are normal: the number of different words (about 2,900 per 10,000 words, as in medieval technical texts), Zipf’s law, and the apparently weak “syntax” (which is the same as in Latin technical texts: it was the Bible, a very formulaic text, that was the wrong benchmark).

4.2 The space is a weak boundary inside a chain of glyphs

This is the first new finding, and it is the key to many of the others.

- **The space can be guessed from the neighbouring glyphs.** Looking only at the two glyphs before and the two after a point, one predicts whether there is a space with an accuracy (F1) of 0.86, higher than in 97% of languages (median 0.64). The spaces that transcribers mark as uncertain fall exactly where the rule is uncertain, and where ZL and Takahashi disagree.
- **The space does not depend on the word.** In languages, when a space is “optional” (for instance in a compound), the choice depends strongly on the whole word. In the Voynich it does not: only the neighbouring glyphs and the length of the piece being written matter (the scribe breaks a long piece more often and avoids detaching pieces of 1–3 glyphs).
- **Different cuts yield real words.** Removing the spaces and putting them back at random with the rule of the neighbouring glyphs, half of the wrongly cut pieces are nevertheless words that exist in the Voynich (50%), against 6.5% in languages (maximum 25%).
- **The vocabulary fills the possible forms.** Among the glyph sequences most probable under the rules of the system, 43% are Voynich words; in languages 4%. And the frequent words are exactly the most probable sequences (correlation 0.58, against 0.11 in languages).
- **The rules between words mirror those inside words.** The preferences between the last glyph of a word and the first of the next resemble those between two consecutive glyphs inside a word (correlation 0.47), more than in all 71 languages (at most 0.39, usually below 0.2). The same rule holds inside words written without a space: internal q ’s follow a glyph $-y$, $-o$ or $-d$ in 28 cases out of 29.

In short, the text behaves as a **chain of glyphs** with its own sequence rules, in which spaces are inserted where the chain allows it. Voynich “words” are not units chosen one by one as in a language; they are pieces of a chain. This does not yet say what the chain represents.

4.3 Junction and junction rules between neighbouring words

- **A strong junction.** The last glyph of a word says a lot about the first glyph of the next: 0.18–0.19 bits beyond chance. Of 71 languages only six exceed the Voynich (among them Sanskrit, which writes the assimilations between words, and the Arabic of the Qur'an). In the gibberish handwritten by volunteers it is 0.014; in published generators between 0 and 0.08.
- **Two precise rules.**
 - *qo-* or *o-* before the gallows: *qo-* after a word ending in *-y*, *-o*, *-d* (59–64% of the time), *o-* after *-n*, *-r*, *-s*, *-m* (*qo-* only 23–27%).
 - *-l* or *-r* at the end of a word: *-r* before a word beginning with *a-* (85%), *-l* before *k-*, *t-*, *d-*, *l-*, *s-*, *q-* (63–86%).
 - They have the shape of rules such as English *a/an* or French *liaison*. They do not say which sounds lie behind them.
- **The scribe looks one word ahead.** When two words copied from the line above would jointly violate a rule, the scribe changes the piece easiest to change: for *-l/-r* the end of the first word (59% against 21%), for *qo-/o-* the start of the second (60% against 18%). So while finishing a word the scribe already knows the next one.
- **And no further.** For a given word in between, the last glyph of a word says nothing about the first glyph of the word after next; in languages this link exists in 34 texts out of 42. The Voynich looks one word ahead and one back, no more.
- **A rule of the system.** The rules hold equally in the three main hands, in all sections, in all bifolia, and with all three transliterations.

4.4 The line is a closed unit

- **The junction stops at the line break.** Between the last word of a line and the first of the line below the link is zero, against 0.19 inside the line. In the prose of languages, which breaks lines wherever it happens, the link carries over by half or entirely; it is close to zero only in texts in which each line is a verse.
- **And at the gap of a drawing.** Where a line is interrupted by a plant, the link between the words on the two sides drops from 0.167 to 0.026. In a language the grammar would carry over the drawing. Words after the drawing begin like words at the start of a line.
- **Line-edge forms.** For the same rest of the word, at the end of a line *-m* rises by 15 percentage points and *-r* falls by 12 (*dar* in mid-line, *dam* at line end); at the start of a line *s-* and *y-* rise by about 9–10 points in place of *ch-* and *k-*. In the last line of a paragraph, which does not reach the margin, *-m* is rarer: it is a form tied to the margin.
- **With no preceding word, *o-* is written.** In the labels of drawings, which are isolated words, *qo-* before the gallows is almost absent (0–5% against 30–68% in lines): *qo-* arises only after a word written continuously. This was a prediction made in advance and confirmed on unseen data.

No published generator has both a chain with weak spaces and a closed line: generator U3 has the chain but lets it cross the line break; the others have neither.

5. The page and the book: copying from the line above, avoiding repetition

5.1 The scribe copies from the line immediately above

- **Copying.** A word has an identical or nearly identical word in the two lines immediately above far more than chance ($z = 17.4$ for the same paragraph vocabulary), more than twice as much as in the generator of Timm and Schinner, which was built precisely on this idea.
- **Only the line immediately above.** Copying comes from the line just above (strongly) and a little from the one before; from the third line up, nothing. In languages the “recurrence” of words is spread over 3–4 lines (it is the continuity of discourse, with a maximum at two lines); in the Voynich it is entirely at distance one. The Timm and Schinner generator draws from up to six lines above: this is the first clear difference with their theory.
- **Pieces of line.** Two neighbouring copied words come, more than chance, from two neighbouring words of the line above, usually in the same order but also inverted.
- **Not by columns.** The copied word does not sit in the same column as its source (the apparent alignment came from the line edges).
- **It stops at the paragraph and the page.** The first line of a paragraph does not copy the last line of the previous paragraph, nor the first line of a page the last line of the previous page. Inside the paragraph, from the third line on, copying is constant.
- **The copy adapts to the junction rules.** A word copied from the line above takes the form (*qo-* or *o-*, *-l* or *-r*) that agrees with its *new* neighbour more than that of its source (effect $+0.49$ against $+0.16$). The junction rules are applied at the moment of writing.
- **Gibberish does not do it.** The meaningless text handwritten by volunteers does not copy from the lines above (-0.001); languages do, but with recurrence spread over several lines.

5.2 Each line avoids starting like the line above

- Shuffling the order of lines within paragraphs shows that the scribe **avoids** beginning a line with the same start as the previous line: *qo-* ($z = -9.7$), *o-* (-7.0), *d-* (-3.8), *ch-* (-3.6), *y-* (-3.5). If the line above starts with *qo-*, the line below takes *o-* (the probability of *qo-* drops by 52 points).
- It concerns the start of the line, not any visual edge: where lines resume to the right of a drawing, one under the other, the effect is absent. The line two above does not matter, nor the end of the line above, nor the right margin.
- Ratio of observed to expected identical starts: Voynich 0.51; languages median 0.95 (only 3 of 57 below the Voynich); handwritten gibberish 1.32; generators 0.93–2.05.

5.3 Paragraphs, pages, bifolia

- **The paragraph opens with a large sign.** 83% of paragraph-initial words begin with a gallows (*k*, *t*, *p*, *f* or compounds), against 9% elsewhere; among them *p* dominates (54%), which is rare in the rest of the text. The opening gallows is three times less tied to its word than usual: it is largely a mark of position.
- **First and last lines have their own register.** The first line of a paragraph has longer words, more *p*, *f*, *sh*, more final *-y*; the last has fewer *qo-* and fewer gallows, more *ch-*, *s-*, *y-*.

- **-ey increases down the page** (+0.12 to +0.14 from top to bottom, for the same position in the line), especially in the recipe pages. Lines get shorter down the page.
- **Each page has a strong identity**, mainly along an axis from words in *-edy/-eeey* to words in *-aiin/-ain* (the page departs from the profile of its section 2.8 times more than chance).
- **The bifolium is a working unit.** The identity of a page is shared by the two sides of the same leaf and by the conjugate leaf of the same bifolium (which in the bound book lies far away), not by the facing page. The two halves of a bifolium share rare words and copy from each other. All 48 bifolia with enough data are entirely in language A or entirely in language B, and 94% are in a single hand.
- **Languages A and B differ in vocabulary**, not in glyphs: the best substitution of glyphs brings A closer to B by only 1% of the gap, while the same search recovers a random key 100% of the time. B is not A enciphered with another key.
- **The order of the pages largely preserves the order of writing:** consecutive pages share more words than any two pages of the same section.

6. Choices between variants: repeating, persisting, drifting

In the Voynich many words exist in pairs of variants differing by one glyph: *qo/o* at the start of a word, *k/t* (two gallows), *sh/ch*, *-ey/-dy* at the end. Studying **how the scribe chooses** between these variants turned out to be the most fruitful way of looking at the text.

6.1 Repeating immediately

- In the Voynich 1.0% of adjacent word pairs are the same word repeated (*chol chol*, *qokeedy qokeedy*) and 3.9% are a nearly identical word (one glyph different). In languages: 0.10% and 0.23% (medians; maxima 0.71% and 2.2%); in the six Nordic scribes 0.01–0.05% and 0.2–0.5%; in the 73 German scribes median 0.05% and 0.46% (maxima 0.54% and 1.20%); in handwritten medieval technical texts (herbals, recipe books, astrology) 0.03–0.15% and 0.2–0.6%; in the Copiale copyist practically never.
- Repetitions are the same in all sections (herbal 1.0%, biological 1.2%, pharmaceutical 0.9%, recipes 0.9%): they do not depend on the genre of the content.
- **Repetition fades quickly:** within the line it is strong between nearby words and disappears after 6–7 words (far/near ratio 0.12). In languages it is the opposite (median 2.86): languages avoid repeating at once, the Voynich repeats at once. No comparison text has this signature.
- When a repeated word changes its start or end, the change follows the junction rule with its neighbour (*okeey qokeey*: after *-y* a *q* is added).

6.2 The “state” of the choices

This is the finding that required most work, and its description was refined several times along the way (Appendix B). This is the final state.

- **The short state.** Between nearby words of the same line the scribe tends to make the same choice. Removing the preferences of each word, of the page and of the position, and correcting the bias of the measure, the agreement is +0.13 between adjacent words and

+0.09 at two and three words (*k/t*, *sh/ch*, *-ey/-dy* together; +0.14 for *k/t* and +0.18 for *-ey/-dy* alone). It halves in about three words.

- **It is not copying.** Agreement is almost the same between very different words (four or more glyphs apart: +0.09) and between similar words (+0.13). The Timm and Schinner generator, which copies nearby words with modifications, does the opposite: agreement only between similar words.
- **It is not position in the line.** With a comparison that preserves position (and a “placebo” to separate the effect of the procedure), the part due to position is about zero.
- **It is separate for each choice.** Between different choices the agreement is zero (−0.01): there is no common “marked mode”; *k/t* and *-ey/-dy* change each on its own.
- **It does not depend on how many times the choice occurs in between:** it lives along the line, from word to word, and is “read” when the choice occurs.
- **It is the same in all hands** (hands 1, 2, 3: between +0.12 and +0.15 between adjacent words), in languages A and B, in all sections where it can be measured, and in the three transliterations (in Glen Claston the gallows +0.12, the endings +0.14). It is a **rule of the writing system**, not the habit of one scribe.
- **Variants of form alone show no clear agreement.** The two forms of *d* and of *e* that only Glen Claston distinguishes (differences in the shape of the sign, not of the word) have a weak and uncertain agreement (+0.05 / +0.06, intervals touching zero). We do not draw a general rule from this: a real scribe has a window in a choice of form alone (Section 7).
- **A slow state, in addition.** Besides the short state there is a slow component that crosses the line break: words of consecutive lines agree (+0.05), at two lines less (+0.03), at 7–12 lines not at all. It is not a top-to-bottom trend of the page. Unlike the short component, this slow component is **normal in real scribes** (Section 7).

What it says and what it does not. A state lasting a few words, separate for each choice, is compatible with a regulated writing habit (the writer “enters” a variant and stays in it for a while), with a rule of an encryption system, or with a procedure that produces text. It does not say that the variants carry information, nor which.

6.3 Drift along the line

- Inside the line, for the same word and without the first and last word, the variants *qo*, *k*, *sh*, *-ey* become rarer towards the right: from −0.020 to −0.034 per 10 glyphs, that is 8–12 percentage points from one end of a typical line to the other. The drift is steeper in the first half of the line and is present in every hand, about three times stronger in language B.
- Being common to all choices, it is not an effect of ink on a single stroke (for example the small stroke that distinguishes *sh* from *ch*). No published generator has it.
- Next to drawings the marked forms become rarer; in particular the *q* drops in the word that begins right after a drawing (−33 points), not in the one before. Without the images we cannot tell whether this is writing (restarting against the drawing with *o-*) or reading by the transcribers.

7. Comparison with people who actually wrote by hand

Many of the properties above could, in principle, be normal habits of a scribe: people writing by hand abbreviate more at the end of a line, alternate letter forms, change habits from one day to the next. A systematic comparison with real scribes was missing, because it requires transcriptions that preserve letter forms and the original lines and pages. We found them in two public archives: Menota (Nordic manuscripts) and the ReF (German manuscripts). To these we added the copyist of the Copiale cipher, who chooses between equivalent symbols as someone writing in a cipher might.

For each property we used **the same measure** as for the Voynich, recomputing the Voynich in the same run.

Table 1: The “scribe battery”: properties of the Voynich measured identically on real scribes and on the Copiale copyist. Values in brackets are pending (experiment e3c77 on the German scribes).

property	Voynich	6 Nordic scribes (1200–1550)	73 German scribes (1350–1500)	Copiale copyist	verdict
junction rule (effect / uncertainty of the choice)	0.058	2 of 11 choices at 0.013–0.015, others zero	2 of 13 at 0.001–0.005, others zero	—	proper to the Voynich (4–12 times the strongest scribe)
left margin: lines avoiding the start of the line above	5 initials avoided	0 of 72 tests	no manuscript with ≥ 3 initials avoided; 3 of 72 with one	—	proper to the Voynich
line-edge forms (rise of the most typical form)	end $-m$ +0.151; start $s-$ +0.098	+0.016–0.025 (abbreviations at line end, $v-$ at start)	+0.019 (line-break hyphen); +0.012	—	same nature, 4–8 times stronger in the Voynich
repetition of the adjacent word (identical / near)	1.00% / 3.87%	0.01–0.05% / 0.2–0.5%	median 0.05% / 0.46%	0.00% / 0.04%	proper to the Voynich
drift along the line (per 10 glyphs)	−0.020 to −0.034	at most 0.0075; abbreviation +0.005	[e3c77]	—	proper to the Voynich (so far)
short state of choices (adjacent; 2–3 words)	+0.13; +0.09, equal for different words	1 of 18 choices with a similar window, but coming from similar words	[e3c77]	rotates homophones: −0.18	not found in scribes in the Voynich’s form
slow state between consecutive lines	+0.050	6 of 17 choices agree across lines, 5 as large as the Voynich	—	—	normal for handwriting

7.1 What it means

- **Five properties out of seven have no counterpart** in 79 scribes of three centuries and four traditions (Icelandic, Norwegian, Swedish, German), nor in the copyist of a cipher: junction rules, left margin, repetitions and drift along the line (the last so far measured only)

in the six Nordic scribes); line-edge forms have the same nature (a real scribe abbreviates more at line end) but are four to eight times weaker.

- **The Voynich’s short state** (agreement that holds even between very different words) was found in none of the scribes measured so far (the six Nordic scribes; the German ones are in progress). One Norwegian scribe (AM 302 fol, around 1300) has, in the choice between long and round *s*, a “window” of similar size and shape, which passes the same tests as the Voynich (it does not come from short stretches of the page nor from position); but his window comes mainly from **similar words** (+0.29 between similar words, +0.065 between different words), that is, from repeating forms of the same word: a different mechanism.
- **The Copiale copyist does the opposite of the Voynich:** if in the previous word he used one symbol for a letter, in the next word he tends to use another (agreement -0.18 between adjacent words, down to -0.48 for *u*). It is the practice of someone who encrypts with homophones and rotates them so as not to leave repetitions. The Voynich choices do not behave like the homophones of this cipher.
- **The slow state** that carries from one line to the next is instead **normal** for handwriting (habits that change slowly, or changes of hand not marked in the files). It does not distinguish the Voynich.
- **Repetitions do not come from genre either:** fifteenth-century German herbals, recipe books and astrology texts written by hand repeat the adjacent word 7 to 30 times less than the Voynich.

In one sentence: **the “way of writing” of the Voynich is not that of a scribe copying a text in a language with ordinary variant forms.** Its rules between neighbouring words and at line edges are much stronger. This does not say what the text is; it says that whoever wrote it followed rules that ordinary scribes did not follow.

8. What the Voynich is not

Every “no” in this section holds because the method used says “yes” when the answer is there (positive control). Where a control did not work, the test was not used.

8.1 Not a language written in an ordinary way

- The statistical fingerprint (Section 4.1) matches none of the roughly one hundred languages of the sample, nor medieval technical texts.
- **The space is not lexical** (Section 4.2): in languages the space depends on the word, here on the neighbouring glyphs.
- **Immediate repetition and its shape** (Section 6.1): languages avoid repeating at once, the Voynich repeats at once.
- **No link beyond the adjacent word** (Section 4.3), whereas languages have links at a distance as well.
- **An honest caveat:** with the corrected measure, the agreement between Voynich choices has the same size as grammatical agreement in languages (for example between the endings *-o/-a* in Italian or *-us/-a* in Latin), and its shape (how long it lasts at two-three words) lies at the top of the range of languages, not outside it: the technical Latin of Pliny comes close. The short state alone is not enough to distinguish the Voynich from a language with agreement; the other properties together are.

8.2 Not a real text encoded in a simple way

We encoded real texts (Latin, Italian, the Bible, technical texts) in many ways and measured whether they acquire the properties of the Voynich:

- **word-for-word code** (each word replaced by a Voynich word): it takes the letters and vocabulary of the Voynich, but not the repetitions, page homogeneity or junction rules;
- **verbose ciphers** (each letter becomes a group of 2–3 glyphs): of 450 variants none reaches the predictability of the Voynich with words of its length;
- **the Naibbe cipher**: similar in predictability and word length, not in repetitions, unique words, page homogeneity, junction (almost zero) and line properties;
- **abbreviations, null letters, transpositions**: they move away or do not suffice;
- **lists, prayers, litanies** encoded: they take on the “grain” of the Voynich but not its anomalies.

8.3 Not a substitution readable in a known language

- **Homophonic substitution with spaces in 14 languages**: in the controls (the same kind of cipher on another biblical text) 69–98% of real words are recovered; on the Voynich at most 36%, with 7–93 distinct words, mostly always the same three, as for a text in another language.
- **A solver without spaces** (simulated annealing with a 5-gram model of Latin trained on 9 million letters), calibrated on 56 + 71 positive controls (it recovers the key at 98–100%); on the Voynich, counting glyphs in four different ways and at nine levels of grouping, in 71 languages, even reading backwards, the result never reliably exceeds that of the negative controls. The “deciphered” text is a mush of syllables.
- **Other tests**: 85 languages with spaces; 16 vernaculars, historical languages and abbreviated Latin; a mixed nomenclator; searches guided by the expected content (plant names, months); other reading orders; pieces of words as symbols: no reading. Two “formally positive” outcomes (a German candidate and a syllabic one) proved to be artefacts: the second came out identical on the shuffled Voynich.
- **Ordered anagrams**: the order of glyphs inside words is language-like, not anagram-like.
- **Plant names as cribs**: the name of the plant, enciphered on purpose in a control, is recovered in the first word of the page; in the Voynich, with the available identifications, there is no trace.

8.4 The published readings we could test

- **Bax (2014)**: the recurring words do not behave as names (*shor*, read as “black”, occurs 96 times on 67 pages of all sections); outside the pages it was derived from, the key does not beat chance ($p = 0.14$) and produces “names” everywhere.
- **Vatne (2021)**: it relies on its own transcription; only a third of its entries are found in the reference transliteration.
- **Cheshire (2019, “proto-Romance” language)**: his key reads more Romance words than chance, but *any* key of the same form adapted to the manuscript in a few minutes does better, even on pages not used to adapt it (20 keys out of 20).

- **Schechter (2026, glossary from EVA to Latin, Occitan and Hebrew;** according to the author a bilingual Latin–Occitan pharmaceutical manual, “87.8% decoded”): his program, re-implemented, gives the same figures; but a “glossary” made of the most frequent words without any meaning covers as much, his glossary “reads” 67–70% of a text without a message as well, and the order of the decoded words is not Latin (attested pairs as in shuffled text).
- **Gatta (2026, a mapping of 19 EVA glyphs to Hebrew consonants, read right to left):** the signal present on the Voynich is present on the text without a message too, and a mapping searched for on purpose does better than his. The author himself, in the current version of his work, concludes that the text does not behave like Hebrew and that no page is readable.

The **message-free text control** (applying the same reading to a text generated without content) is simple, and we propose it as a minimum test for anyone announcing a decipherment.

8.5 None of the simulated historical cipher mechanisms

We looked in the Voynich for the signature that some fifteenth- and sixteenth-century encryption systems would leave, after checking on texts enciphered on purpose that the test can see it:

- **homophones chosen by the copyist** (as in the Copiale): they would leave a *negative* agreement between nearby words (the copyist alternates the symbols); the Voynich has a positive agreement;
- **a sign that changes the alphabet** (the index letters of Leon Battista Alberti’s cipher disk, *De componendis cifris*, c. 1466): no frequent word or sign after which the state of the choices restarts (a rare index cannot be excluded);
- **a bit message in the binary choices** (like Francis Bacon’s biliteral cipher, devised c. 1576–79 and described in 1623, where the form of the letter carries a bit): a message restarting at each page or line would be seen with overwhelming evidence in the controls (*z* up to 70); in the Voynich there is no signal. A continuous message crossing pages remains outside the reach of the test;
- **word tables in turn** (the “Ave Maria” of Johannes Trithemius’ *Polygraphia*, printed in 1518, where each letter becomes a Latin word taken from the next table and the ciphertext reads like a prayer): no rhythm from two to six words along the line;
- **an autokey restarting at each line** (each encrypted letter depends on the previous one; each line has a new starting key): it is the only mechanism tested that **closes the line** like the Voynich, but it destroys everything else: no repetition, an inflated vocabulary (4,500 to 6,400 distinct words per 10,000, against 2,900), weak line starts, no avoidance at the margin.

8.6 Not a constructed language, not gibberish, not the published generators

- **Constructed languages** (Esperanto, Volapük, Neo-Quenya, Interlingua, Lojban, Klingon, LOLCat): none has the state of the choices; none has even an agreement between adjacent words.
- **Meaningless text handwritten by volunteers:** it has no junction (0.014), no line-edge forms, no copying from the line above, no avoidance of repeated starts (it repeats them). Someone inventing a text by hand does not spontaneously produce the rules of the Voynich.

- **Published generators:** Naibbe has neither junction nor line properties; the Timm and Schinner generator has copying from the lines above and a non-lexical space, but it draws from too far, has no junction closed within the line, no state of the choices (its agreement comes from copying similar words), no margin avoidance and no drift; generators U2 and U3 lack the combination of chain and closed line. **None reproduces together** a junction closed within the line, short memory of repetitions, the state of the choices and margin avoidance.

9. What we confirm and what we refute in previous work

previous idea or result	our outcome
Currier’s languages A and B	Confirmed (98.5% of pages; confirmed by the vocabulary in 107 of 107 herbal pages). They differ in vocabulary, not in glyphs; each bifolium is entirely A or entirely B.
Several hands (Currier; Davis 2020)	Not questioned , but the fine habits distinguishing hands are partly confounded with sections (hands 2 and 3 cannot be distinguished within the herbal). Above all, all follow the same rules (junction, state of the choices, margin).
Self-citation (Timm and Schinner 2020): the scribe copies and modifies words already written	Partly confirmed. The scribe does copy from the line above (more than their generator) and repeats at once. But Voynich reuse is short-range (the line immediately above; in-line memory of 6–7 words), adapted to the junction rules , with a state of the choices that holds between different words: things their generator does not do.
The Naibbe cipher (Greshko 2025) as an explanation	Not sufficient as a complete explanation: no junction, no line properties, repetitions and unique words far off.
Topic “keywords” (Montemurro and Zanette 2013)	Not distinctive: texts generated without a message have them too.
Gibberish resembles the Voynich (Gaskell and Bower 2022, on low-level statistics)	Not in the rules of writing: handwritten gibberish has no junction, no line-edge forms, no copying from the line above, no margin avoidance, nor the regularities of the vocabulary (frequent words are not the most probable forms).
“A glyph is not a letter, a token is not a word, a space is not a space” (Rozanova and Temerev 2026)	Convergent: we too find a chain of glyphs with rules at word edges, glyphs too regular for a one-to-one substitution, and uncertain spaces that behave like word-internal junctures. Like them, we find that the “words” form a plausible vocabulary and that the anomaly lies in how they follow one another.
Readings by Bax, Vatne, Cheshire, Schechter, Gatta	Do not survive the controls (Section 8.4).
Zodiac labels as numbers or days	No: almost all different, no positional correspondence between months.
“The Voynich has little syntax”	Not an anomaly: Latin technical texts have as little.

10. Discussion

10.1 What we know, in one page

The Voynich text is produced by a writing system with three levels of rules:

1. **Inside the word and between neighbouring words:** a chain of glyphs with its own sequence rules, in which spaces are weak boundaries placed where the chain allows; junction rules between the end of a word and the start of the next, applied while writing and looking one word ahead.

2. **Within the line:** the line is a closed unit with its own starting and ending forms; choices between variants follow a short state (two to three words) and drift towards simpler forms from left to right; words are repeated at once and the repetition fades within a few words.
3. **Between lines and across the book:** the line below copies from the line immediately above, adapting the copies to the junction rules, and avoids starting like it; the paragraph has its own opening and closing; page, bifolium and quire have an identity; the bifolium is a working unit.

Many of these rules belong to the **system**, not to the person: they hold equally across hands, across Currier's two "languages", in all sections and in all three transliterations.

10.2 What this excludes and what it leaves open

It excludes, with the controls described:

- a natural language written in an ordinary alphabet, read with a simple or homophonic substitution in one of the languages tested;
- the published readings we could test;
- the simulated historical encryption mechanisms, taken one at a time;
- a text invented by hand without rules (the volunteers' gibberish does not have the Voynich's rules);
- that the Voynich's rules are the ordinary habits of a scribe (79 real scribes do not have them).

It leaves open three scenarios, which our data cannot separate:

1. **A language written with a very artificial writing system:** for example a system in which the variants (*k/t*, *-ey/-dy*, ...) carry no meaning but follow rules of form, and in which spaces do not separate the words of the underlying text. A reading would be possible only with an external anchor.
2. **A code or cipher of an untested kind**, for example with a word repertory or a lost key book. In this case statistics is not enough.
3. **No message:** a text produced by a procedure executable by hand. A procedure of this kind (the self-citation generator of Timm and Schinner, with a junction rule added by us) already reproduces many basic properties (predictability, similarity between words of the page, repetitions, link between neighbouring words), but not the line and page rules described here.

Our data shift the weight slightly towards scenarios 1 and 3 compared with a classical historical cipher, because the Voynich's rules are "rules of form" and no cipher we tested reproduces them. But they **do not separate** them.

10.3 A caveat: a message can hide in the choices

In related work (a tool we call the "voynichizer", developed in parallel), we verified that a real text, compressed and encrypted with a key, can be hidden in the choices between variants (*k/t*, *sh/ch*, *-ey/-dy*, *qo/o*, ...) of a text with the properties of the Voynich, **without making it more recognisable** to a statistical discriminator. The capacity of this channel is about 0.8 bits per choice, that is about 46,000 bits for a book like the Voynich (the equivalent of about

2,000 compressed Latin words). Therefore **statistical properties alone cannot prove that the text has no meaning**: a text “without a message” and a text with a message hidden in its choices can have the same statistics.

10.4 Why statistics alone will not lead to a reading

All our tools measure **regularities**. A reading requires knowing what the signs **represent**, and regularities do not say this: a very regular writing system can hide a language, a code or nothing. A reading would need an external element: a text of which the Voynich is a copy (a known herbal or treatise), a secure identification of many plants to use as cribs, a key found in an archive.

10.5 What would change our mind

- A reading system that passes the controls used here: one that works on the real text and not on the shuffled or message-free text, that produces meaningful sentences on pages never seen before, and that explains the writing rules described (junction rules, closed line, state of the choices, copying from the line above, margin).
- A medieval scribe, or a historical cipher, with the same rules: it would change the reading of the whole work.
- A simple procedure that reproduces all of them together.

11. Limitations and declared gaps

- **Small effects.** Many of the new properties are small in absolute terms (agreement of choices is about 13 percentage points between adjacent words) and required methodological corrections. We report them with intervals and replications, but they remain subtle measurements.
- **Transliterations.** Everything depends on transliterations: the three used disagree on about one word in eight. The main measures hold on all three, but without the images we cannot separate what the scribe did from what the transcribers read (for example the *q* dropping after a drawing).
- **Comparison scribes.** 79 Nordic and German scribes; Latin or Italian fifteenth-century scribes transcribed at facsimile level are missing, as are medieval English scribes (known for very variable spelling). The Nordic and German manuscripts do not mark changes of hand.
- **Historical ciphers.** Only one real cipher (the Copiale, eighteenth century). The DE-CODE archive, which collects hundreds of fifteenth- and sixteenth-century ciphers, requires an account and has no reuse licence; no fifteenth-century cipher appears there with both transcription and key.
- **Tests we could not run**, because the measures, tested on synthetic texts, did not distinguish the cases:
 - whether the state of the choices lasts a number of **words** or a span of **writing time** (glyphs);
 - whether the states of different choices **change at the same points** (which would indicate hidden units of a few words);
 - whether the **short** state crosses the line break (the slow one does).

- **A test with few data:** whether the state survives the gap of a drawing (the estimate is positive but the interval is too wide to tell).
- **In progress** at the time of this draft: the short state and the drift in the 73 German scribes (experiment e3c77); a comparison between manuscripts and printed books of the same corpus.

12. Reproducibility

Each experiment has:

- a dated **preregistration** (`preregistrazioni/eNN.md`), written before running it;
- a **program** (`esperimenti/eNN_*.py`) in Python 3.12, with fixed random seeds;
- an **execution record** with the version of code and data (`risultati/provenienza/`);
- a **result** as a table (`risultati/eNN_*.md`) and in full (`.json`);
- an entry in the **laboratory notebook**, written only by adding.

Re-running an experiment gives the same numbers. The repository is currently private; copyright-protected data are not included and must be downloaded again from the sources (Appendix D).

A. Experiments behind the main findings

Experiment codes refer to the repository (files `preregistrazioni/`, `esperimenti/`, `risultati/`).

finding	main experiments	replications
Statistical fingerprint, isolation from languages	e01–e06, e13	IT, GC
Weak and predictable space, cuts yielding words, vocabulary filling the forms	e3a49, e3a55, e3a58, e3a61, e3a67, e3a99–e3b05	e3a50, e3a56, e3a60, e3a68
Junction and junction rules	e377, e380, e390, e3a01–e3a03, e3a41	e388, e395, e3a04, e3a05, e3a34
Closed line, drawing gap, line-edge forms	e384, e386, e387, e389, e3a09	e395, e3a43
Copying from the line immediately above	e340, e343, e347, e372, e385, e392, e394, e396, e3b28	e3a80, e3a84
Left-margin avoidance	e3a25–e3a27, e3a33, e3a35, e3a44	e3a33 (IT, languages A and B)
Paragraph, page, bifolium, quire, languages A and B	e302, e307–e316, e321, e350, e356, e378	e3a15, e3a21
Immediate repetition and its shape	e02, e3a86–e3a90, e3b25	e3a88, e3a91, e3a92
Short state of the choices	e3c33, e3c48, e3c52, e3c55, e3c57, e3c66, e3c72, e3c81	IT, GC (e3c51), hands, sections (e3c54)
Slow state	e3c73–e3c75	e3c76 (scribes)
Drift along the line	e3c13, e3c17, e3c20–e3c24	IT
Scribe battery	e3c50, e3c58–e3c65, e3c76–e3c80	—
Copiale	e3c61	—
Decipherments and readings excluded	e14, e16–e20, e26, e27, e133, e163, e163b, e212b, e216, e219, e222, e223, e249, e269	positive controls in every test
Historical cipher mechanisms	e3c68–e3c71	synthetic controls in every test
Constructed languages, gibberish, generators	e346, e374, e377, e382, e399, e3a95, e3c37, e3c49, e3c53	—

B. Internal checks: provisional indications corrected by the method

None of the results of this work had been published before. During the work, however, some intermediate measurements gave indications that later checks corrected. We report them because they show which measurements are fragile and how the controls corrected them: anyone redoing the work can avoid the same missteps, and readers can judge how far the final conclusions have been tested.

Corrections due to the bias of the agreement measure (Section 3.3).

1. A first measurement attributed to the Anglo-Saxon scribe of the *Hatton Gospels* the same “window” as the Voynich. For that choice the measure was inflated (+0.25); corrected, the scribe has agreement between adjacent words but not at two-three words.
2. Volapük seemed to have an agreement similar to the Voynich; with the corrected measure it is zero.
3. The shape of the Voynich’s agreement (how long it lasts at two-three words) seemed outside the range of languages; corrected, it lies at the top of the range, not outside.
4. An observation about “neutral” words between two words making the same choice suggested that the state travels only on occurrences of the choice; it was an effect of the same bias.

Descriptions made more precise by later measurements.

5. The state of the choices looked like a “window” of three words then dropping to zero; instead it fades gradually (half-life about three words) and there is in addition a slow component carrying from one line to the next.
6. The hypothesis that each line has its own “setting” of the choices (like a key per line) did not hold: agreement carries from one line to the next.
7. After the test on Glen Claston’s form-only variants we had read the state as tied to “which word is written, not how the letter is drawn”; a real scribe with a window in a form-only choice showed that the generalisation was too broad (and his window, in turn, comes from similar words).
8. An uncorrected measure indicated that hand 1 lacked the state of the choices; with the corrected measure all hands have it equally.
9. A measure indicated that the memory of the choices resets at the line break; it was an effect of comparing with the page average (line ends and starts have their own forms).
10. The first left-margin test used a defective comparison; with the correct comparison the result holds.
11. A measure attributed two thirds of the small link between whole words to the paragraph; it was a mechanical effect of smaller groups.
12. Rare forms seemed to be copied from the line above more than others; with the second transliteration only a trend remains.

Formally positive outcomes that were artefacts.

13. A “one word = one syllable” reading candidate and a German candidate in a content-guided search: the first came out identical on the shuffled Voynich, the second changed sign when only the starting points of the search were changed.

Criteria phrased too sharply.

14. In three cases the criterion decided in advance called “absent” or “reset” an effect whose uncertainty interval touched zero (the state in the herbal, at the drawing gap, in some sections). In all three the estimate was positive and similar to the rest: these were insufficient data, not absence. Hence the rule, which we suggest to others too, of saying “absent” only when the interval lies entirely below the reference value.

C. Glossary

EVA conventional alphabet to transliterate the Voynich into Latin letters; it does not indicate sounds.

glyph a sign of the manuscript; *ch*, *sh*, the gallows and their compounds count as one.

gallows the tall signs *k*, *t*, *p*, *f* of EVA.

Currier languages A and B the two varieties of writing into which the pages divide.

junction how much information the last glyph of a word gives about the first glyph of the next, beyond chance.

junction rule a rule linking the form of a word to that of its neighbour (sandhi-like).

closed line the junction does not carry over the line break.

state of the choices the tendency to repeat the same variant for a few words.

in-line drift the decline of marked variants from left to right.

preregistration a written description, before running, of question, measure and criterion.

positive / negative control a text in which the answer is / is not present.

homophonic cipher the same letter can be written with several symbols.

verbose cipher each letter becomes a group of several glyphs.

self-citation a procedure in which the writer copies and modifies words already written.

facsimile-level transcription a transcription preserving the letter forms of the manuscript.

95% interval the range in which the true value lies, with good confidence.

D. Sources and licences

D.1 Data

The Voynich.

- R. Zandbergen, transliteration files of the Voynich manuscript in IVTFF format, <https://www.voyrich.nu/transcr.html>; Zandbergen–Landini (ZL3b, EVA), Takahashi (IT2a), Glen Claston (GC2a, v101 alphabet); format described in R. Zandbergen, *IVTFF – Intermediate Voynich MS Transliteration File Format*, v2.0.1 (2025).
- Radiocarbon dating of the vellum (1404–1438, 95%): NSF-Arizona AMS Laboratory, University of Arizona, team led by G. Hodgins, four samples measured in 2009; University of Arizona press release, 10 February 2011; account by R. Zandbergen, <https://www.voyrich.nu/extra/carbon.html>.

Comparison texts.

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- D. E. Gaskell and C. L. Bower (2022), “Gibberish after all? Voynichese is statistically similar to human-produced samples of meaningless text”, in C. Layfield and J. Abela (eds.), *Proceedings of the 1st International Conference on the Voynich Manuscript 2022 (VOY2022)*, CEUR Workshop Proceedings 3313, paper 4 (2023), <https://ceur-ws.org/Vol-3313/paper4.pdf>. The meaningless texts were handwritten by volunteers (42 in the study; 38 texts in the published data); data licence: modified MIT.
- M. A. Greshko (2025), “The Naibbe cipher: a substitution cipher that encrypts Latin and Italian as Voynich Manuscript-like ciphertext”, *Cryptologia*, published online 26 November 2025, doi:10.1080/01611194.2025.2566408 (code: `greshko/naibbe-cipher`).

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- **Menota**, Medieval Nordic Text Archive, <https://www.menota.org> (catalogue on Clarino, University of Bergen). The licence is stated text by text; for the six texts used (AM 519 a 4to, AM 677 4to, AM 60 4to, AM 242 fol, Holm A 10, AM 302 fol) it is CC BY-SA 4.0. Each text should be cited with the data of its header (editor, version).
- **ReF**: K.-P. Wegera, H.-J. Solms, U. Demske, S. Dipper (2021), *Reference Corpus of Early New High German (1350–1650)*, version 1.0.2, Zenodo, doi:10.5281/zenodo.5793616. Zenodo states CC BY 4.0, the LICENSE file inside the archive CC BY-SA 4.0: we follow the more restrictive.
- **Copiale**: K. Knight, B. Megyesi, C. Schaefer (2011), “The Copiale Cipher”, in *Proceedings of the 4th Workshop on Building and Using Comparable Corpora (BUCC)*, ACL, pp. 2–9, <https://aclanthology.org/W11-1202/>; and K. Knight, B. Megyesi, C. Schaefer, “The Secrets of the Copiale Cipher”, *Journal for Research into Freemasonry and Fraternalism* 2(2): 314–324 (2011 volume, published 2012), doi:10.1558/jrff.v2i2.314. Transcription and decipherment from the project page of Stockholm University. Dating: mid-eighteenth century (the inscription “1866” in the manuscript is an owner’s mark).

D.2 Readings and decipherments tested

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